American International University-Bangladesh (AIUB)
Department of Computer Science Faculty of Science & Technology (FST)

PROJECT TITLE

MediAssist (AI-Powered Online Pharmacy System)

A Software Engineering Project Submitted
By

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1. PROJECT PROPOSAL

1.1 Background to the Problem

Problem Domain

Accessing medicines and basic pharmaceutical advice is a basic need for health care. In many developed nations, this necessity becomes fulfilled progressively by digital pharmacy platforms which allow patients to order medicines online, communicate to pharmacists from a distance, and get it delivered to their door. But in Bangladesh, the pharmaceutical industry is still mostly offline, which makes it hard for millions of people to get the medicines they need safely and quickly.

Bangladesh has more than 170 million people, but a lot of them, especially those who live in rural and semi-urban areas, don't have easy access to qualified pharmacists or well-stocked pharmacies. Most pharmacies are in big cities like Dhaka and other divisional headquarters. However, even these city pharmacies don't always have all the medicines that people need, so they have to go to more than one store, and sometimes they can't find the medicine at all. Also, these city pharmacies are often too full, don't always have the right stock, and don't have any digital integration. Patients often go to more than one pharmacy to get the same medicine. This issue is much worse in rural areas, where there are few qualified pharmacists and pharmacies that have enough supplies. There is no reliable digital platform where a patient can describe their symptoms and get AI-assisted medicine suggestions before talking to a pharmacist. This is the most important thing. We often see a dangerous cretaria where people get self-medication without professional guidance. In the absence of qualified pharmacist consultation, people in rural areas often take medicines based on their neighbors' advice, incomplete knowledge, or outdated prescriptions. As a result of wrong dosages, people get sick more often, and their health worsens.

The idea of creating Medi-Assist is to solve those problems directly. By creating a web-based platform that combines an AI-powered symptom checker with mandatory pharmacist verification, the system provides a safe, structured, and accessible path from symptom to medicine without requiring the patient to physically visit a pharmacy for every minor ailment. The AI handles initial guidance, the pharmacist handles verification, and the delivery person handles the delivery side. This three-layer model ensures both convenience and medical safety.

Why This System is Needed Now

The demand for digital healthcare services in Bangladesh has improved significantly since the COVID-19 pandemic, which forced millions of people to seek remote alternatives for basic medical needs. Medi-Assist addresses this gap by improving the most common everyday healthcare interaction, buying medicine, and making it safer, faster, and more accessible through technology. It bridges the disconnect between patients and qualified pharmacists, particularly for those in rural and semi-urban areas who currently have no reliable way to verify medications, check stock availability, or receive professional guidance before making a purchase.

Target Group of Users

Medi-Assist is designed for four distinct user groups, each of them have specific needs and responsibilities within the system:

Customer

Customers are everyday people, mostly patients,or family members who need to buy medicines without going to a physical pharmacy. They could be elderly people who are unable to go physically, working people who do not have time during the day, or people living in areas where pharmacies are far away or out of stock.

Responsibilities: Customers search for medicines, use the AI symptom checker to get medicine suggestions, upload prescriptions when required by the system, place orders, and track their delivery in real time.

Pharmacist

Pharmacists are licensed medical professionals registered on the platform. They are the human safety layer of the system. Before any AI-suggested medicine reaches the customer, the pharmacist reviews it to make sure it is medically appropriate. They also verify uploaded prescriptions and manage the medicine inventory.

Responsibilities: Pharmacists log in to their dashboard, review AI-generated medicine suggestions, approve or reject customer prescriptions, confirm or cancel orders, and keep medicine stock updated with low-stock alerts.

Delivery Person

Delivery persons are the field staff responsible for picking up confirmed orders from the pharmacy and delivering them to the customer's address.

Responsibilities: Delivery persons log in, view their assigned orders, check customer addresses and contact details, update the delivery status at each stage from pickup to handover, and mark orders as delivered once completed.

Administrator

The administrator is the platform manager who keeps everything running behind the scenes. They do not sell medicines or deliver orders. Their job is to make sure every other role on the platform is working correctly, accounts are properly managed, and the system is secure and up to date.

Responsibilities: Administrators manage all user accounts across every role, monitor orders and AI recommendation logs, oversee the medicine catalog, generate system reports, handle complaints, and configure system settings and AI parameters.